

Private Geometric Median

Joint work with Jon Ullman and Thomas Steinke

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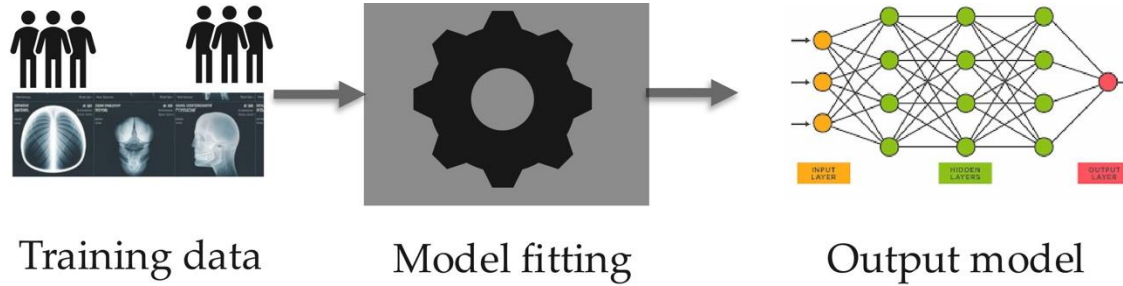
Toyota Technological Institute at Chicago

Private Optimization

Loss function f : parameters $\times$ data $\rightarrow \mathbb{R}$

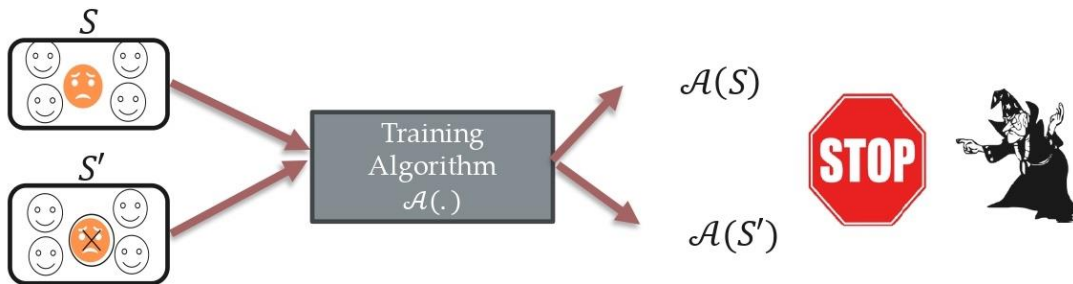
Training set with sensitive information of people

$$S_n = \{(X_i, Y_i)_{i \in [n]}\}$$



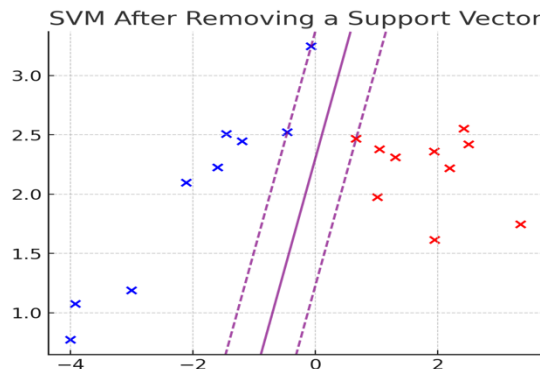
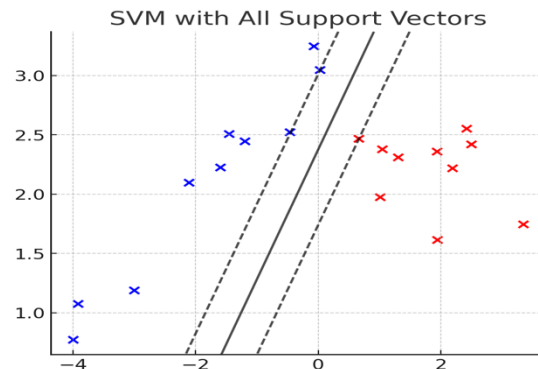
Example: logistic regression

$$F(\theta; S_n) = \frac{1}{n} \sum_{i=1}^n \log(1 + \exp(-\langle \theta, x_i y_i \rangle))$$



Dwork-McSherry-Nissim-Smith 06

A training algorithm $\mathcal{A}$ is (ϵ, δ) -DP, if for every dataset S and S' differ in one sample for every output B

$$\Pr(\mathcal{A}(S) \in B) \leq e^\epsilon \Pr(\mathcal{A}(S') \in B) + \delta$$


DP-SGD Algorithm

1. Initialization Point $\theta_0 \in \Theta$
2. For T iterations
 1. Sample $\xi_t \sim N(0, \sigma^2 I_d)$
 2. Query gradient $\nabla F(\theta_t; X^n)$
 3. perform $\theta_{t+1} \leftarrow \Pi_{\Theta}(\theta_t - \eta(\nabla F(\theta_t; X^n) + \xi_t))$



google-deepmind/
jax_privacy

Algorithms for Privacy-Preserving Machine
Learning in JAX



Theorem [Bassily-Smith-Thakurta 2014]:

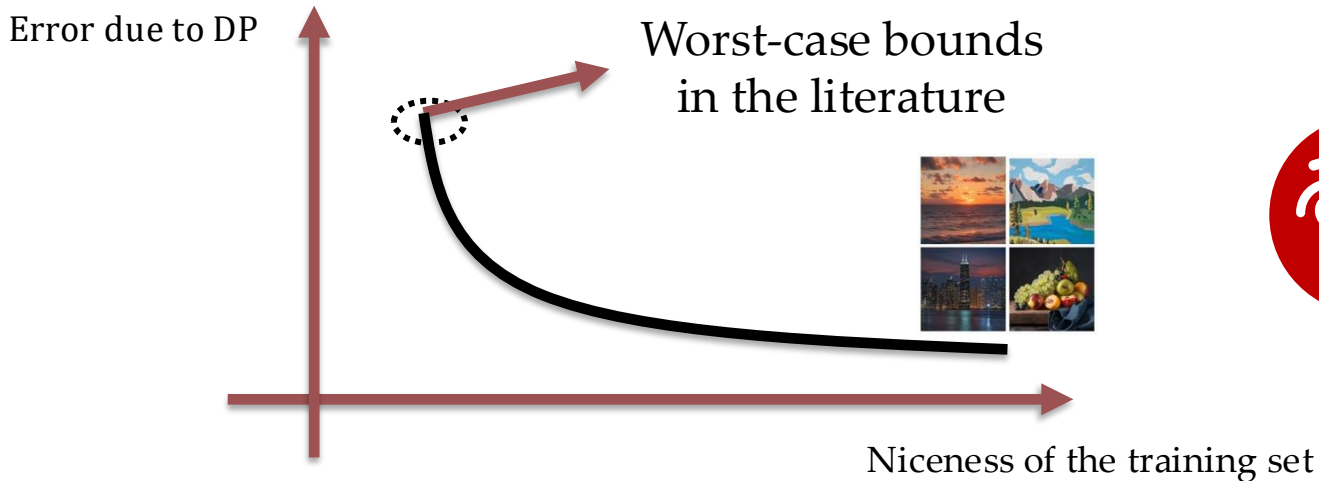
For every z , let $f(\cdot, z)$ be a convex function that is **L-Lipschitz** defined over the ball of **radius R** in $\mathbb{R}^d$. Then, DP-GD algorithm (with specific parameters) satisfies (ϵ, δ) DP. Does this always

Are we done here?
No!

Moreover, there exists a loss function $f(\cdot, z)$ of **radius R** in $\mathbb{R}^d$ and there exists a training data such that

$$\text{excess optimization error due to DP} \geq \frac{LR \sqrt{d \log(\frac{1}{\delta})}}{\epsilon n}$$

Worst-Case Privacy, Data-Dependent Utility



DP requires worst-case privacy guarantees.

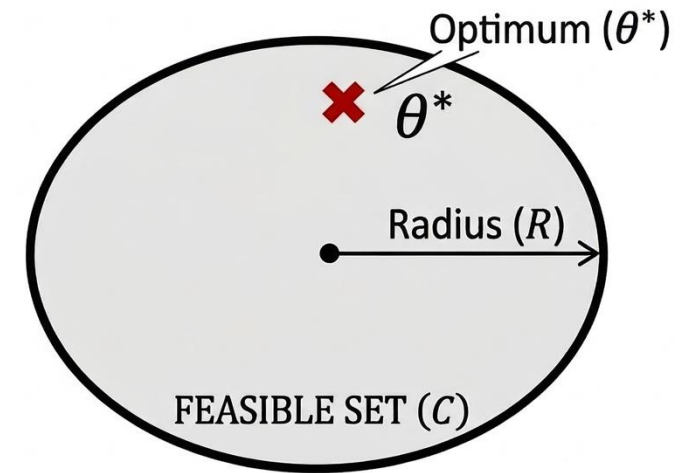
Goal:
Worst-Case Privacy, Data-Dependent Utility

- Lots of success in learning with DP.
- Example: Gaussian mean estimation
 - Without any assumption, private mean estimation of unbounded data is not possible. [Bun-Ullman-Vadhan'14, Steinke-Ullman'15]
 - If the data is Gaussian, much stronger results are possible. [KV17, KLSU19, BDKU20, BGS+21, LKJO22, KDH23, BHS23,...]

Unknown Search Radius

DP-GD pays linearly for uncertainty about where the optimum is.

- We only know that θ^* lies in a large ball of radius R .
- The error due to DP of DP-GD scales linearly with R .
- We can't search forever due to privacy budget.



Main Question:

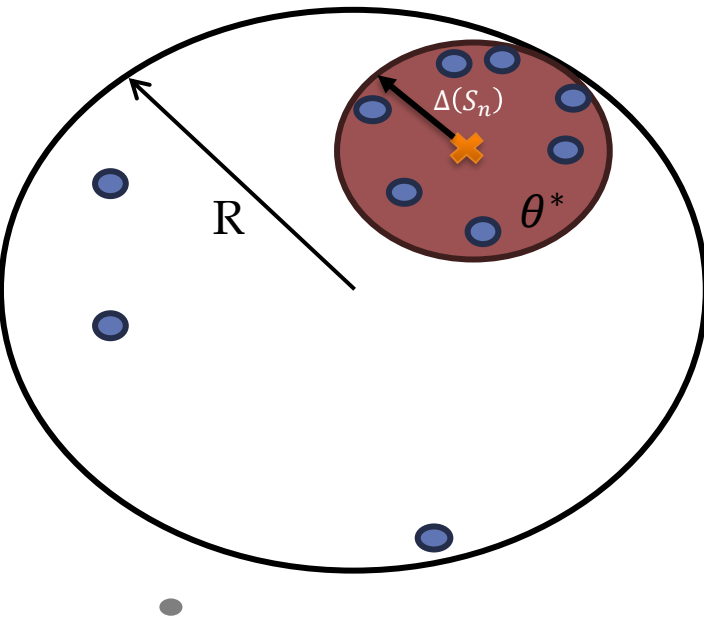
Can we replace this global radius R by a more local, data-dependent notion of complexity of search space?

The Sandbox: Private Geometric Median

- Given $X^n = (x_1, \dots, x_n) \in (\mathbb{R}^d)^n$ such that $\|x_i\|_2 \leq R$, find a private approximate minimizer of

$$F(\theta; X^n) = \sum \| \theta - x_i \|_2$$

- The Geometric Median is a fundamental robust estimator: **ignores outliers**.
 - Subroutine in robust aggregation in FL, confidence boosting, clustering,



Baseline: DP-(S)GD: $\mathbb{E}_{\hat{\theta} \sim \mathcal{A}(X^n)} [F(\hat{\theta}; S_n)] - \min_{\theta \in \Theta} F(\theta; S_n) \leq \frac{R\sqrt{d}}{\epsilon}$

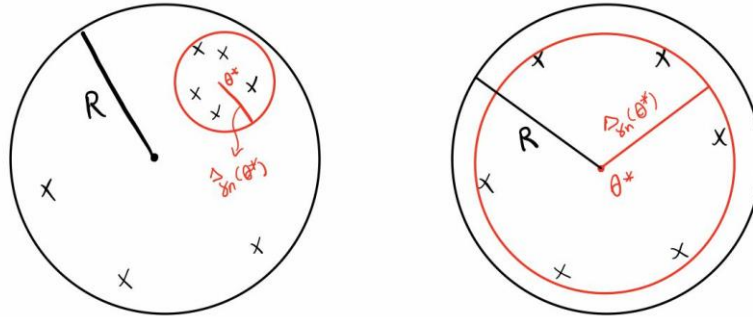
Challenge

DP-SGD pays a worst-case penalty R that breaks this natural robustness.

The Goal: Escaping the Worst-Case Bound

Definition (Quantile Radius):

Fix a dataset $X^n = (x_1, \dots, x_n) \in (\mathbb{R}^d)^n$ and let θ^* be the geometric median of X^n . Then, for every $\gamma \in [0,1]$, define $\Delta_{\gamma n}(\theta^*) = \min\{\Delta: |\{i \in [n]: \|\theta^* - x_i\|_2 \leq \Delta\}| \geq \gamma n\}$



$$\mathbb{E}_{\hat{\theta} \sim \text{DPGD}(X^n)}[F(\hat{\theta}, X^n)] - \min_{\theta \in \Theta} F(\theta, X^n) \leq \frac{R \sqrt{d \log(\frac{1}{\delta})}}{\varepsilon}$$



$$\mathbb{E}_{\hat{\theta} \sim \mathcal{A}(X^n)}[F(\hat{\theta}, X^n)] - \min_{\theta \in \Theta} F(\theta, X^n) \leq \frac{\Delta_{0.8n}(\theta^*) \sqrt{d \log(\frac{1}{\delta})}}{\varepsilon}$$

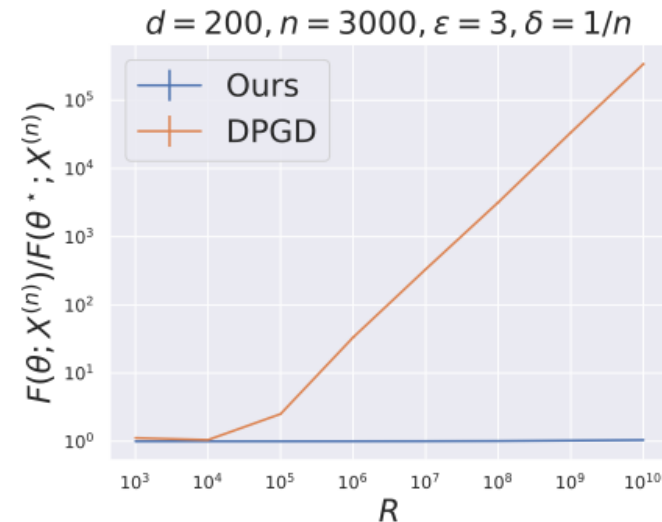
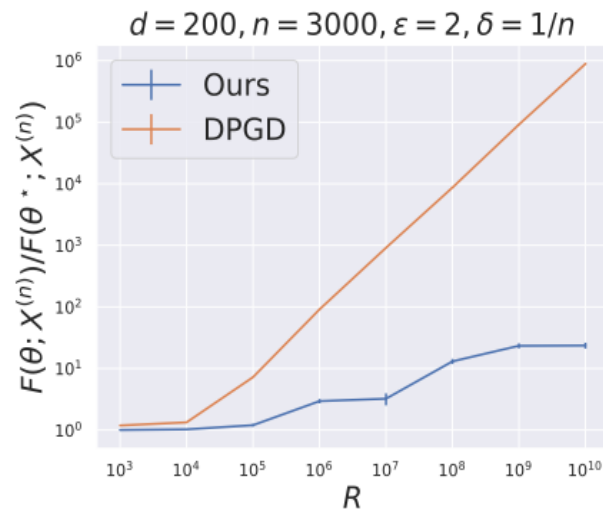
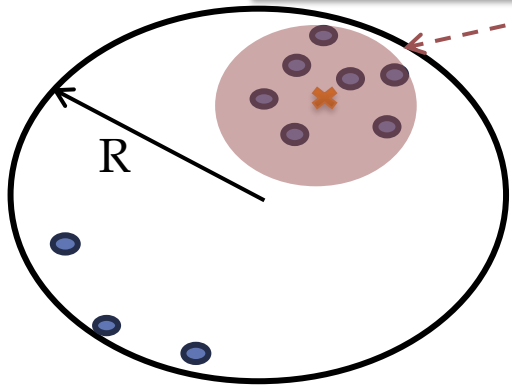
Main Result: Polynomial-Time Algorithm

Theorem [Haghifam-Steinke-Ullman NeurIPS'24]:

There exists a polynomial time (ε, δ) DP algorithm such that with a high prob

$$F(\hat{\theta}; X^n) - F(\theta^*; X^n) \leq \frac{\text{(quantile radius)} \sqrt{d}}{\varepsilon}$$

Moreover, the sample complexity $\frac{\sqrt{d \log R}}{\varepsilon}$ and runtime of $n^2(d + \log(R))$.



Recent work of Kumar-Liu-Tian-Yang '25 improves the runtime to linear.

Baseline: DP-(S)GD: $\mathbb{E}_{\hat{\theta} \sim \mathcal{A}(X^n)} [F(\hat{\theta}; X^n)] - \min_{\theta \in \Theta} F(\theta; X^n) \leq \frac{R\sqrt{d}}{\varepsilon}$

This result requires an additional assumption: assume there exists a very small known constant $r > 0$ such that $\Delta_{0.8n}(\theta^*) > r$.

Blow up in the sample complexity is $\log\left(\frac{1}{r}\right)$

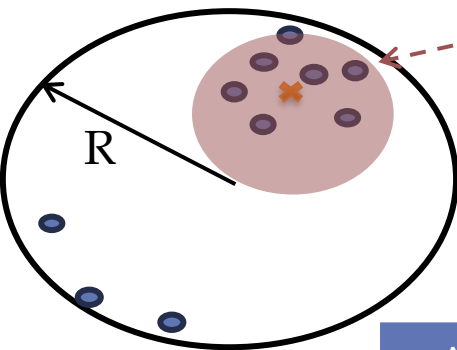
Main Result: Inefficient Pure DP

Theorem [Haghifam-Steinke-Ullman NeurIPS'24]:

There exists an inefficient $(\varepsilon, 0)$ –DP algorithm such that with a high prob

$$F(\hat{\theta}; X^n) - F(\theta^*; X^n) \leq \frac{\text{(quantile radius)} d}{\varepsilon}$$

with the sample complexity $\frac{d \log(R)}{\varepsilon}$.



This result requires an additional assumption: assume there exists a very small known constant $r > 0$ such that $\Delta_{0.8n}(\theta^*) > r$.
Blow up in the sample complexity is $\log\left(\frac{1}{r}\right)$

Algorithms	Function Error	Sample Complexity	Runtime
Approx-DP	$F(\hat{\theta}; X^n) \leq \left(1 + O\left(\frac{\sqrt{d}}{n \varepsilon}\right)\right) OPT$	$\frac{\sqrt{d \log(R)}}{\varepsilon}$	$n^2 d$
Pure-DP	$F(\hat{\theta}; X^n) \leq \left(1 + O\left(\frac{d \log(R)}{n \varepsilon}\right)\right) OPT$	$\frac{d \log(R)}{\varepsilon}$	Exponential

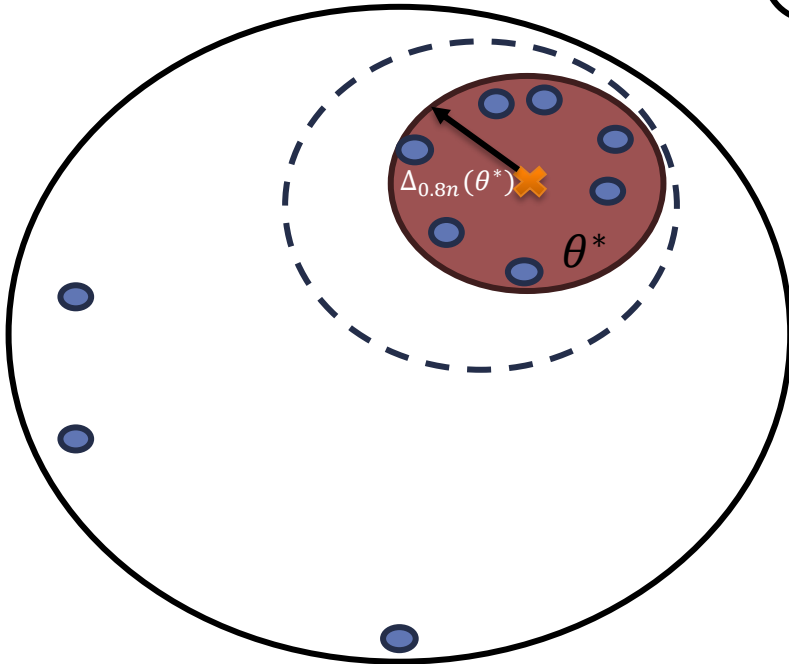
Outline of the Talk

- Motivation and Problem Statement
- Main Results
- Some ideas behind the algorithms

What doesn't feel right about DP-GD?

Our Goal:

$$\mathbb{E}_{\hat{\theta} \sim \mathcal{A}(X^n)}[F(\hat{\theta}, X^n)] - \min_{\theta \in \Theta} F(\theta, X^n) \leq \frac{\Delta_{0.8n}(\theta^*) \sqrt{d \log(\frac{1}{\delta})}}{\varepsilon}$$



Large Stepsize → **Instability** around the optimal point.
Small Stepsize → Cannot reach the optimum because of the **privacy cost**.

Private Localization

Private Localization (Informal)

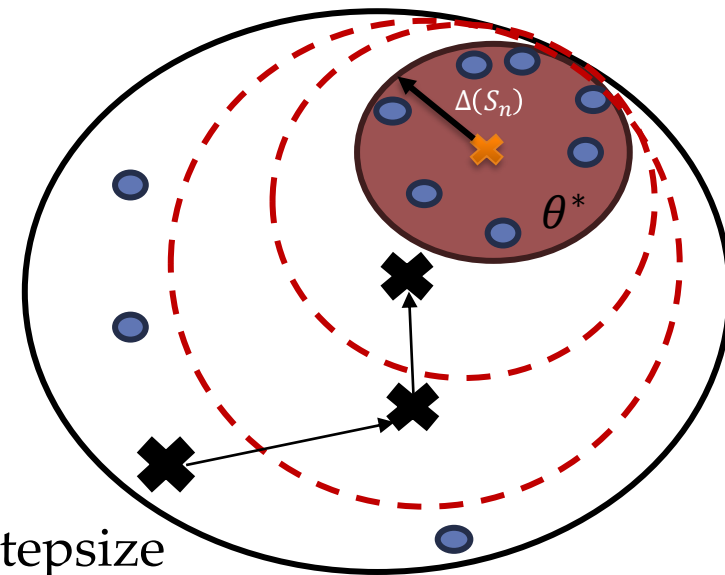
- Compute $\hat{\Delta}$ a private estimate of $\Delta_{0.8n}(S_n)$
- Number of phases $k = \log(\frac{R}{\hat{\Delta}})$
- $\text{rad}_0 = R$
- For $t \in \{1, \dots, k\}$:
 - $\eta_t \propto \text{rad}_t$
 - $\theta_{t+1} = \text{DPGD}(\theta_t, \Theta_t, \eta_t, 500)$
 - $\text{rad}_{t+1} = \frac{1}{2} \text{rad}_t$
 - $\Theta_{t+1} = \{\theta: \|\theta - \theta_{t+1}\| \leq \text{rad}_{t+1}\}$

DP estimate of
the effective size

Start with Large and
geometrically decaying stepsize

Run DPGD for constant iterations

Update the feasible set: new feasible set is smaller!



Lemma (Informal):

In $\log(R)$ iterations, with a high probability, the **updated feasible set** satisfies

1- It contains the exact geometric median.

2- It has a radius proportional to $\Delta_{0.8n}(\theta^*)$. (**not** with the apriori worst-case bound R .)

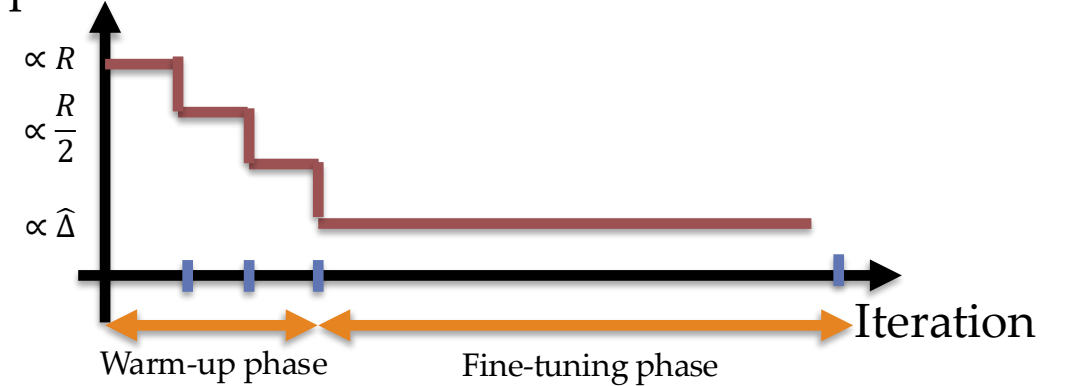
Fine-Tuning

After the localization step, we have a good initialization point and a smaller feasible set with radius $O(\Delta_{0.6n}(\theta^*))$.

Algorithm (Informal)

- $(\theta_0, \hat{\Delta}) = \text{warmup}(X^n)$
- $\Theta_{\text{loc}} = \{\theta: \|\theta - \theta_0\| \leq \hat{\Delta}\}$
- $\eta \propto \hat{\Delta}$
- $\hat{\theta} = \text{DPGD}(\theta_0, X^{(n)}, \Theta_{\text{loc}}, \eta, n^2)$

Stepsize schedule



Theorem (Informal):

Given $n \geq \Omega\left(\frac{\sqrt{d \log R}}{\epsilon}\right)$, with a high probability, we have

$$F(\hat{\theta}, X^n) - \min_{\theta \in \Theta} F(\theta, X^n) \leq \frac{\Delta_{0.6n}(\theta^*) \sqrt{d \log(\frac{1}{\delta})}}{\epsilon}$$

Why does it work?

When the current point is far from θ^* , it should be “safe” to take a bigger gradient steps.

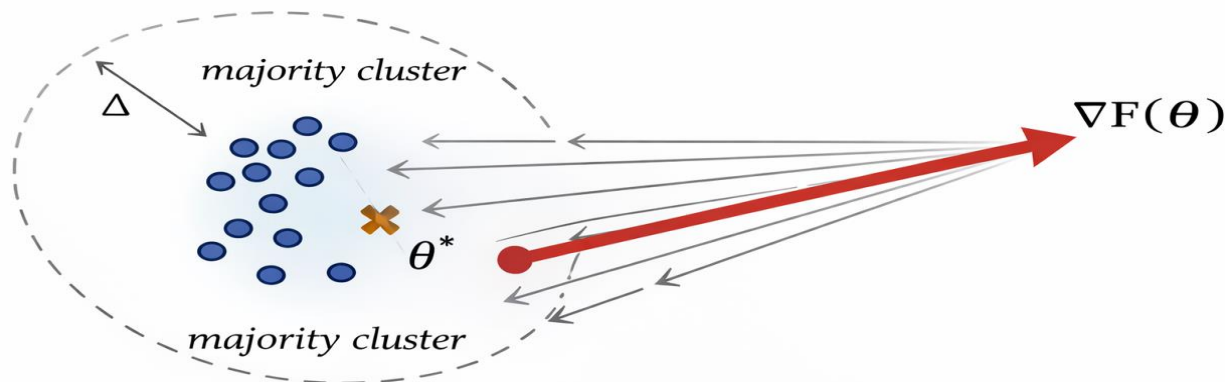
Gradient of GM is Robust to Outliers

$$F(\theta; S_n) = \sum \|\theta - x_i\|_2 \quad \longrightarrow \quad \nabla \|\theta - x_i\| = \begin{cases} \frac{\theta - x_i}{\|\theta - x_i\|}, & \text{if } \|\theta - x_i\| > 0 \\ \text{arbitrary unit vector,} & \text{otherwise} \end{cases}$$

Lemma (Informal)

For a given θ such that $\|\theta - \theta^*\| \geq C \cdot \Delta_{0.8n}(\theta^*)$, we have a linear growth:

$$\frac{F(\theta; S_n) - F(\theta^*; S_n)}{n} \geq \|\theta - \theta^*\|$$



Warmup Method

$$\|\theta - \theta^*\| \leq \frac{F(\theta; \mathbf{X}^{(n)}) - F(\theta^*; \mathbf{X}^{(n)})}{0.2n} + 6\Delta_{0.6n}(\theta^*)$$

$\Theta_0 = B_d(R)$ $\longrightarrow$ Run DPGD initialized at θ_0 for T iterations and call the output θ_1

$$F(\theta_1; \mathbf{X}^{(n)}) - F(\theta^*; \mathbf{X}^{(n)}) \leq \|\theta_0 - \theta^*\| \left(\frac{\sqrt{d}}{\epsilon n} + \frac{1}{\sqrt{T}} \right) \leq \frac{\|\theta_0 - \theta^*\|}{5} \rightarrow \text{Assuming } n \text{ is large enough by running DP-GD for constant number of steps.}$$

$$\|\theta_1 - \theta^*\| \leq \frac{F(\theta_1; \mathbf{X}^{(n)}) - F(\theta^*; \mathbf{X}^{(n)})}{n} + 6\Delta_{0.6n}(\theta^*) \leq \frac{\|\theta_0 - \theta^*\|}{5} + 6\Delta_{0.6n}(\theta^*)$$

$$\|\theta_1 - \theta^*\| = \frac{2}{5} \|\theta_0 - \theta^*\| \text{ given that } \|\theta_0 - \theta^*\| \geq 30\Delta_{0.6n}(\theta^*)$$

Warmup Method: Finding Quantile Radius

- $\Delta_{\gamma n}(\theta^*)$ is a data-dependent quantity.
- **Intuition:** If a data point is close to many other points then that point needs to be close to GM.
- Let $N_i(v) = |\{k: \|x_i - x_k\| \leq v\}|$ and let $N(v) = \max N_i(v)$

Lemma) if $N(v) > \gamma n$ and $N(v/2) < \gamma n$ then $\frac{2\gamma-1}{4\gamma-1} \Delta_{\gamma n}(\theta^*) < v < 4\Delta_{\gamma n}(\theta^*)$.

- Challenge for privacy: The centers are datapoints!
- An idea from Nissim-Stemmer-Vadhan'16: $N_{stable}(v) = \frac{1}{m} \max_{\{i_1, \dots, i_m\}} (N_{i_1}(v) + \dots + N_{i_m}(v))$

What is the $N_{stable}(v)$ for $v = 4r$?



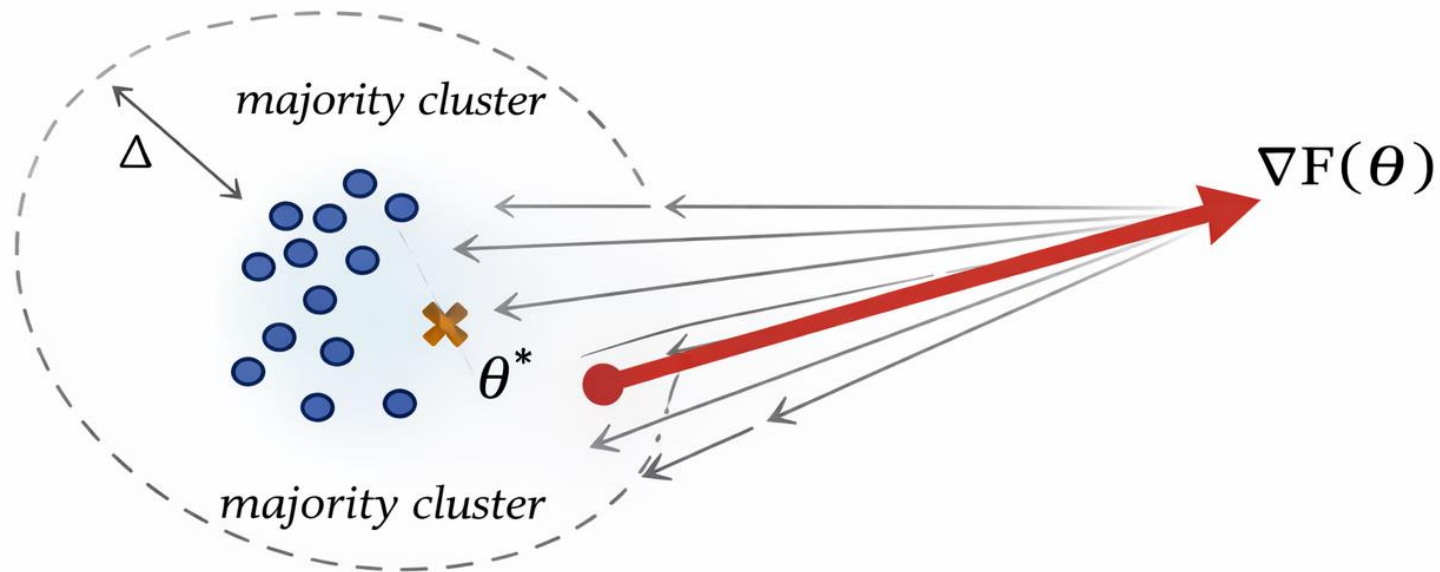
Lemma)

Given $n \geq \frac{1}{\epsilon} \log \left(\frac{R}{r} \right)$, there exists a pure-DP algorithm that outputs $\hat{\Delta}$ such that with a high probability, $c_1 \Delta_{0.75n}(\theta^*) \leq \hat{\Delta} \leq c_2 \Delta_{0.8n}(\theta^*)$

What we have seen so far?

Gradient of GM is Robust to Outliers

$$F(\theta; S_n) = \sum \|\theta - x_i\|_2 \quad \longrightarrow \quad \nabla \|\theta - x_i\| = \begin{cases} \frac{\theta - x_i}{\|\theta - x_i\|}, & \text{if } \|\theta - x_i\| > 0 \\ \text{arbitrary unit vector,} & \text{otherwise} \end{cases}$$



Next I'll show a second algorithm that exploits the same stability from a sampling viewpoint.

Pure DP Algorithm

Geometric median is **stable**

one needs to change many points to significantly move the geometric median

Smooth Inverse Sensitivity (Informal)

- Length function

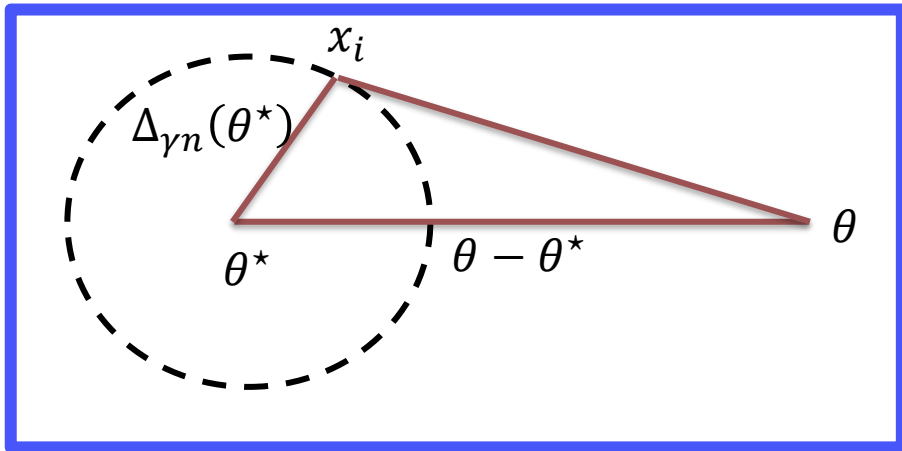
$$\text{len}(y) = \min_{\tilde{X}^n} \{d_H(\tilde{X}^n, X^n) \text{ s.t. } GM(\tilde{X}^n) = y\}$$

- “Compute” $\text{len}(y)$ for every $y \in B_d(R)$
- Sample from $\hat{\theta} \propto \exp\left(-\frac{\epsilon}{2} \text{len}(y)\right)$

Asi-Duchi’20, Asi-Ullman-Zakynthinou’23, Hopkins-Kamath-Majid-Narayanan’23

Why does it work?

- Simple analysis shows that the output of this mechanism can be thought of as the GM of a dataset $\tilde{X}^n$ that differ in at most $O\left(\frac{d \log R}{\varepsilon}\right)$ points from the original dataset X^n .
- Let the remaining points in the ball of radius $\Delta_{\gamma n}(\theta^*)$ denoted by $m = \gamma n - O\left(\frac{d \log R}{\varepsilon}\right)$
- How can we control $\|GM(X^n) - GM(\tilde{X}^n)\|$?



$$\begin{aligned} \|\nabla F(\theta, \tilde{X}^n)\| &\geq \sum \left\langle \nabla \|\theta - \tilde{x}_i\|, \frac{\theta - \theta^*}{\|\theta - \theta^*\|} \right\rangle \\ &\geq \sum_{\substack{i \in \text{remaining point} \\ \text{in the ball}}} \left\langle \frac{\theta - x_i}{\|\theta - x_i\|}, \frac{\theta - \theta^*}{\|\theta - \theta^*\|} \right\rangle - (n - m) \end{aligned}$$

$$\left\langle \frac{\theta - x_i}{\|\theta - x_i\|}, \frac{\theta - \theta^*}{\|\theta - \theta^*\|} \right\rangle \geq \sqrt{1 - \left(\frac{\Delta_{\gamma n}(\theta^*)}{\|\theta - \theta^*\|}\right)^2}$$

Takeaways and Future Work

Takeaway: Private optimization with data adaptive guarantee for GM

- Both algorithm relies on the stability of the gradient of geometric median:
 - For polynomial time algorithm, we show that **gradient stability gives curvature**
 - For pure-DP algorithm, we show that **gradient stability won't let the GM move too much.**

Future Work:

- Extend local, data-dependent guarantees beyond geometric median.